Traffic and Crash Detection with Software Defined Networking

Timothy Marchant
Department of Electrical and Computer Engineering
Kennesaw State University
Marrietta, GA

Aaron Zaman
Department of Electrical and Computer Engineering
Kennesaw State University
Marrietta, GA

Josh Green
Department of Electrical and Computer Engineering
Kennesaw State University
Marrietta, GA

***Abstract*—Traffic accidents account for approximately 1.3 million fatalities globally per year, with survival rates strongly influenced by the speed of emergency response. Existing accident responses rely heavily on human bystander reporting and centralized processing, which can introduce latency, and/or missed reporting. While Software Defined Networking (SDN) has demonstrated performance benefits in urban and highly congested environments, its effectiveness in rural deployments remains less understood. This paper presents a proof-of-concept system that explores the use of SDN for traffic accident detection and response in a distributed setting, scalable for a rural network.**

The proposed architecture is implemented in a Mininet-WiFi virtual machine, modeling a network which includes a wireless traffic camera system, storage, emergency service centers, and induced mobility with roaming emergency vehicles. Performance is evaluated with respect to latency, scalability, and network load. Results indicate that, unlike in urban scenarios, SDN does not provide a clear advantage in reducing latency within the sparse design. These findings suggest that improving network coverage and communication reach may be more impactful than routing optimization alone. This work highlights the importance of environment-specific design when considering applying SDN to a network.

Keywords—mininet-wifi, software defined networking (SDN), traffic monitoring

I. Introduction

Globally, approximately 1.3 million people die in traffic accidents [1]. The odds of surviving a traffic accident improve the sooner help arrives. Commonly, bystanders witness an accident and report it to emergency services. There is several flaws with this and can lead to life threatening latency. Additionally, the phenomenon known as the bystander effect can induce critical latency. The bystander effect is the fact that many people refrain from helping, especially when aware that other people are present at the scene. [2].

There has been much work done in evaluating the use of cameras in traffic. Whether its for speed detection using image processing [3] or evaluating traffic patterns and integrating with the light systems and vehicle access points to reduce congestion [1]. However, there has been very little evaluation on the impact these systems could have in more rural environments. This paper proposes research that seeks to reduce accident reporting latency by offloading the reporting mechanisms to camera systems equipped with a Software Defined Networking (SDN) node to quickly route traffic. Not only would this research be impactful in a rural setting, but could later be integrated into existing research, such as Passable, the integrated system designed for congestion reduction.

II. Model Architecture

For the purposes of this research, we utilized Mininet-WiFi, a fork of the Mininet SDN network emulator. We ran the virtual machine (VM) installation with Oracle VirtualBox. Our source code can be found on GitHub (https://github.com/TimothyMarchant/EE6220FinalProject).

The model was broken down into several components to develop the topology. We identified the need for a traffic node, a middlebox housing the SDN and camera monitoring software, a storage center, an emergency service center, and mobile emergency vehicles.

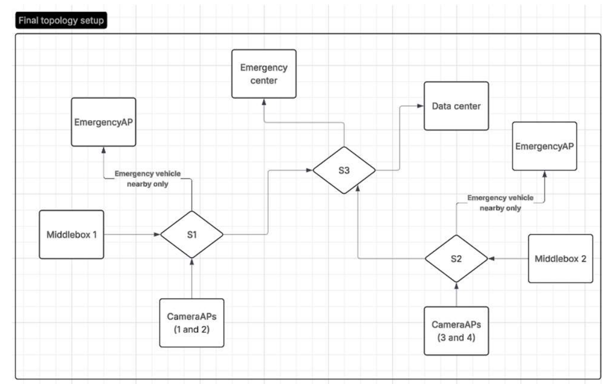

Figure 1:Topology Diagram

The codebase is broken down into two main components. The first component is the Topology, which is a Python file to generate all the entities within the simulation. The entities are then initialized via a series of Python functions that instruct them to send and receive data. These functions exist within the PythonSrc folder. Lastly, a set of OpenFlow scripts are used to route the traffic between the nodes. These OpenFlow controls are where the SDN can modify the flow of network traffic. The entities are described in the following sections.

A. Topology Entities

1) Traffic Node

A traffic node represents a single monitored intersection and serves as the primary data collection point in the system. Each node consists of a traffic camera and a wireless access point. The camera continuously captures traffic data and streams it to a local intersection access point (AP) using wireless communication (Wi-Fi).

The intersection AP maintains a connection to a designated middlebox for image processing. Additionally, the wireless AP is available for connection to emergency service vehicles, providing an alternative path to reduce latency for the middlebox to share emergency information.

Originally to simplify implementation, captured video streams are abstracted as data transmissions equivalent in size to standard image resolutions (e.g. 480p frames) at a variable frame rate. This allows the system to emulate realistic network load without requiring development of video processing capabilities. However, the VM suffered major performance problems when just sending very little information, so to counteract this we sent very short strings in place of this. It only sends data to its corresponding middlebox.

2) Middlebox

The middlebox functions as the primary processing and decision-making unit within the network. We treat it as an abstract "black box" responsible for analyzing incoming traffic data and determining appropriate routing actions. Depending on the deployment scale, a middlebox may be a small, embedded edge device or a massive data center. For the purposes of this study, the behavior is simplified, restricting a middlebox to handling an intersection worth of data at the requisite frame rates.

Additionally, a requirement for data storage has been included in this study. We determined that non-accident traffic data can be stored at a reduced frame rate, preserving storage and network bandwidth for accident data. Not included in this study is the assumption that the middlebox would be used to route other network data. This data would also be required to be deprioritized in the event of emergency detections.

3) Data Center

The datacenter's purpose was to serve as a place to store all the camera information. For this project, its purpose was meant for simulating constant traffic on the lines towards the emergency center.

4) Emergency Service Center

The emergency service center acts as a response center to emergencies. So, if an emergency is detected at a middlebox, the operator can respond to it. It is very important that data be delivered to the emergency center as quickly as possible. This was implemented with a GUI that could respond to these events.

5) Emergency Vehicles

Emergency vehicles were established as stations and provided the mobility aspect of our simulation. We initiated the location of emergency vehicles in a consistent location with a fixed direction. Figure 2 shows the Mininet topology graph.

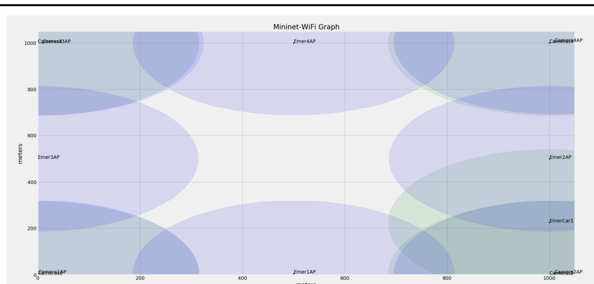

Figure 2: Mininet Topology Graph

B. OpenFlow and SDN

Upon initialization, we establish OpenFlow rules for each of the entities in the simulation. This is with NormalRouterOpenflowCMDs.sh. The SDN controls allow the flow of emergency traffic to be prioritized, and institutes rules for what data can be dropped. The SDN is automated and is ran by SDNFlowCMDs.sh.

III. Simulation and Evaluation

We measured our simulation using throughput at the middleboxes and routing to the Emergency Center. A higher throughput to the Emergency Center from the middlebox indicates the capability to handle additional traffic beyond the baseline. Traffic nodes would likely run four cameras to monitor an intersection. These cameras transmit constant data to the middlebox and emergency service center on a constant flow. When an accident is detected that additional data needs to be stored and streamed to either the emergency center or emergency vehicles, concurrently with the existing baseline stream.

The data is evaluated from the perspective of the Middlebox, where traffic detection occurs and the SDN resides. Figure 3 shows the transmission of data from the middlebox to the emergency center. Figure 4 shows the reception of data from the emergency center. We include a column for Apps, as we developed Python GUI displays to initiate alerts and monitor the network. These apps had a

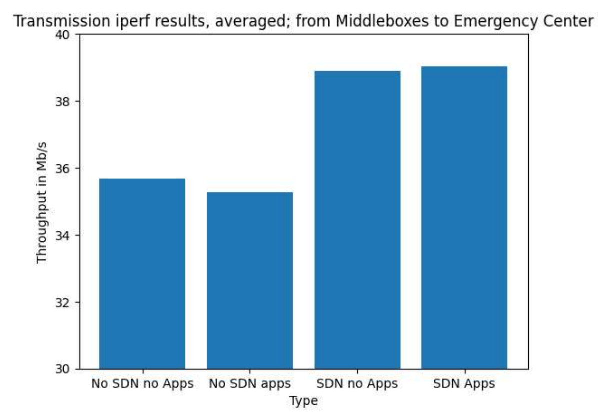

Figure 3: Middlebox Transmission Throughput

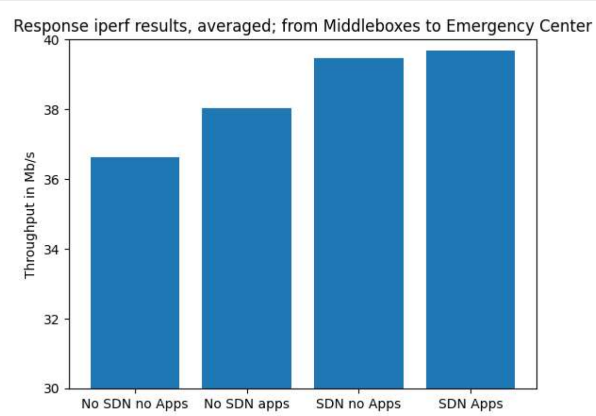

Figure 4: Middlebox Receiving Throughput

performance impact on the network in our early development. We were able to reduce this impact and increase performance.

Without SDN and static OpenFlow matching, we transmitted 35.68 Mbps and received 36.62 Mbps. With the python apps running, we had a noticeable reduction in performance on the middlebox, bringing our transmission rate down to 35.28. It is likely that with a monte carlo run of the simulation, these values would normalize to a marginal impact as observed with the results in comparison between columns 3 and 4 with apps and no apps showing an increase. Next, looking at measurements with SDN, we have 38.9 and 39.04 on transmission and 39.46 and 39.68 receiving, with no apps and with apps respectively.

Averaging our results on transmission and receiving without SDN and compared to the results of transmission and receive with SDN, we see an increase from 36.4 to 39.27 Mbps bandwidth, a 7.8% increase, suggesting the SDN increased the capability of the network.

IV. Conclusion

We successfully demonstrated enhanced routing capability using SDN in our proof-of-concept system. Latency was reduced and response time was enhanced. However, the initial goal of extending SDN controllers to a rural network is likely not as impactful. Rural areas would likely benefit from an increase in network reach over latency reduction.

Future work on this topic would include adding our logic into an existing architecture, such as Passable [1] to combine traffic congestion routing with emergency notification.

We acknowledge Mininet-WiFi impact on our research. Mininet-WiFi proved to be a valuable platform for establishing OpenFlow matching and forwarding rules. However, our topology proved to be too taxing for the virtual machine. We were able to effectively demonstrate an SDN controller's logic receiving alerts and routing traffic and reducing latency, but the simulation was incapable of running efficiently in real-time.

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